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Mini Project Report
On

” Incubator for silkworm ”

Submitted in partial fulfilment for the award of degree of
Bachelor of Engineering

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a bonafide students of 5th Semester Electronics & Communication Engineering, Atria Institute of Technology, Bengaluru in partial fulfillment for the award of Bachelor of Engineering in Visvesvaraya Technological University, Belagavi during the year **2025-26**. It is certified that all corrections/suggestions indicated for Internal Assessment have been incorporated in the Report deposited in the departmental library. The mini project report has been approved as it satisfies the academic requirements in respect of mini project work prescribed for the said Degree.

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We the students of third year, Bachelor of Engineering, Electronics and Communication Engineering, Atria Institute of Technology, Bangalore-560024, hereby declare that the Mini Project work entitled “IOT Based Alcohol and Sleep Detector” has been carried out at ATRIA INSTITUTE OF TECHNOLOGY under the guidance of PROF RAMYA C N, Assistant Professor, Department of Electronics and Communication Engineering, ATRIA INSTITUTE OF TECHNOLOGY, Bangalore.

We declare that the work submitted in this report is our own and has not been previously submitted for the fulfillment of the B.E degree at the Visvesvaraya Technological University, Belagavi or any other Institution/University.

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ABSTRACT

Sericulture requires precise control of environmental factors such as temperature, humidity, and air quality to ensure healthy silkworm growth and high-quality cocoon production. Traditional manual monitoring methods are inefficient, labour-intensive, and unable to consistently maintain the required climatic conditions, often resulting in reduced yield and increased risk of disease. To address this challenge, the proposed project presents an IoT-based Silkworm Incubator System designed using an ESP32 microcontroller integrated with sensors such as DHT11 for temperature and humidity monitoring and MQ135 for detecting harmful gases like CO₂, NH₃, and other pollutants. The system automates climate control using actuators such as exhaust fans, LEDs, buzzers, and relays to maintain optimal rearing conditions and provide real-time alerts when thresholds are exceeded. By enabling continuous monitoring, automated control, and reduced human dependency, the system improves productivity, minimizes crop loss, and ensures a stable and favourable environment for silkworm rearing. This smart incubator provides a cost-effective and efficient solution for modern sericulture farms, supporting better silk quality and higher yield.

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CHAPTER 1

Chapter 1:

INTRODUCTION

In spite of high demand for sericulture and silkworms, maintaining optimal conditions such as temperature and humidity on a farm is difficult through manual efforts alone. Lack of mechanization in sericulture sector is a significant factor contributing to difference in production. The proposed system is an embedded system that would continuously monitor and control the climatic conditions of sericulture farm. And we used to detect and measure the concentration of various air pollutants/gases (e.g., CO₂, NH₃, alcohol). The accumulation of waste products and CO₂ from silkworms can be harmful. The MQ135 monitors air quality, prompting ventilation when pollutant levels exceed safe thresholds. Climatic factors have significant influence on disease incidence in silkworm which leads to crop loss.

Maintaining optimal conditions - temperature and humidity on farm is difficult through manual efforts. IoT to lively monitor environmental changes. More manpower is required to cut the mulberry plants and carry them to the silkworm shed. The above can be concluded from the surveyed papers.

1.1 Problem Statement

Traditional sericulture methods struggle to consistently maintain the precise environmental conditions required for optimal silkworm growth and high-quality cocoon production, leading to inefficiencies, significant resource wastage, and high human dependency. Existing practices are often reactive, lack real-time monitoring of critical parameters like temperature, humidity, and toxic gases, and are thus susceptible to variations that negatively impact yield and sustainability.

1.2. Aim:

The aim of silkworm incubator is to provide stable and favourable conditions for maximum egg hatching and healthy silkworm rearing

1.3. Objectives:

Monitor environmental factors

- To continuously track temperature, humidity, and air quality using sensors (DHT11 and MQ135).
- To detect toxic gases like CO, NH₃, and harmful compounds and ensure safety.
- Automate Climate Control

To activate exhaust fans when temperature exceeds limits

1.4. Literature Survey

“Design and Development of IoT Based Smart Sericulture Plant” IEEE 2024 presents an IoT-enabled system that allows real-time monitoring and remote control of sericulture operations through Wi-Fi connectivity.

Internet of Things–based smart sericulture farm incubator ICECA 2021 designed to provide a cost-effective and accessible solution for sustainable sericulture. Their system automates environmental control processes, reducing the need for human supervision and improving consistency in maintaining ideal rearing conditions.

“Smart Sericulture Monitoring System Using IoT,”IPACT 2023introduces a novel automated monitoring system that utilizes sensors such as DHT11 and MQ135 along with a microcontroller to regulate critical environmental parameters—temperature, humidity, and air quality. The paper reports improved silkworm productivity and silk quality due to stable environmental maintenance, while also significantly reducing manual labor compared to traditional sericulture practices.

CHAPTER 2

Chapter 2: **METHODOLOGY**

2.1 INTRODUCTION:

The methodology of the proposed **IoT-based Silkworm Incubator System** focuses on designing an automated environment-control mechanism that ensures optimal conditions for silkworm rearing. The approach integrates sensor-based monitoring, microcontroller-driven decision making, and automatic actuation of climate-control devices. The system begins by continuously sensing temperature, humidity, and air quality using the DHT11 and MQ135 sensors. These real-time readings are fed into the ESP32 microcontroller, which serves as the central processing unit of the entire incubator setup.

Based on predefined threshold values essential for healthy silkworm development, the ESP32 analyzes the sensor data and activates the corresponding output devices. If temperature or humidity deviates from the ideal range, the microcontroller triggers the exhaust fan through a relay, turns on alert indicators, or activates a buzzer for immediate attention. An LCD display is used to present live environmental parameters, ensuring transparency and ease of monitoring. Additionally, a manual key switch is included to operate external components like the DC motor whenever needed.

2.2 BLOCK DIAGRAM

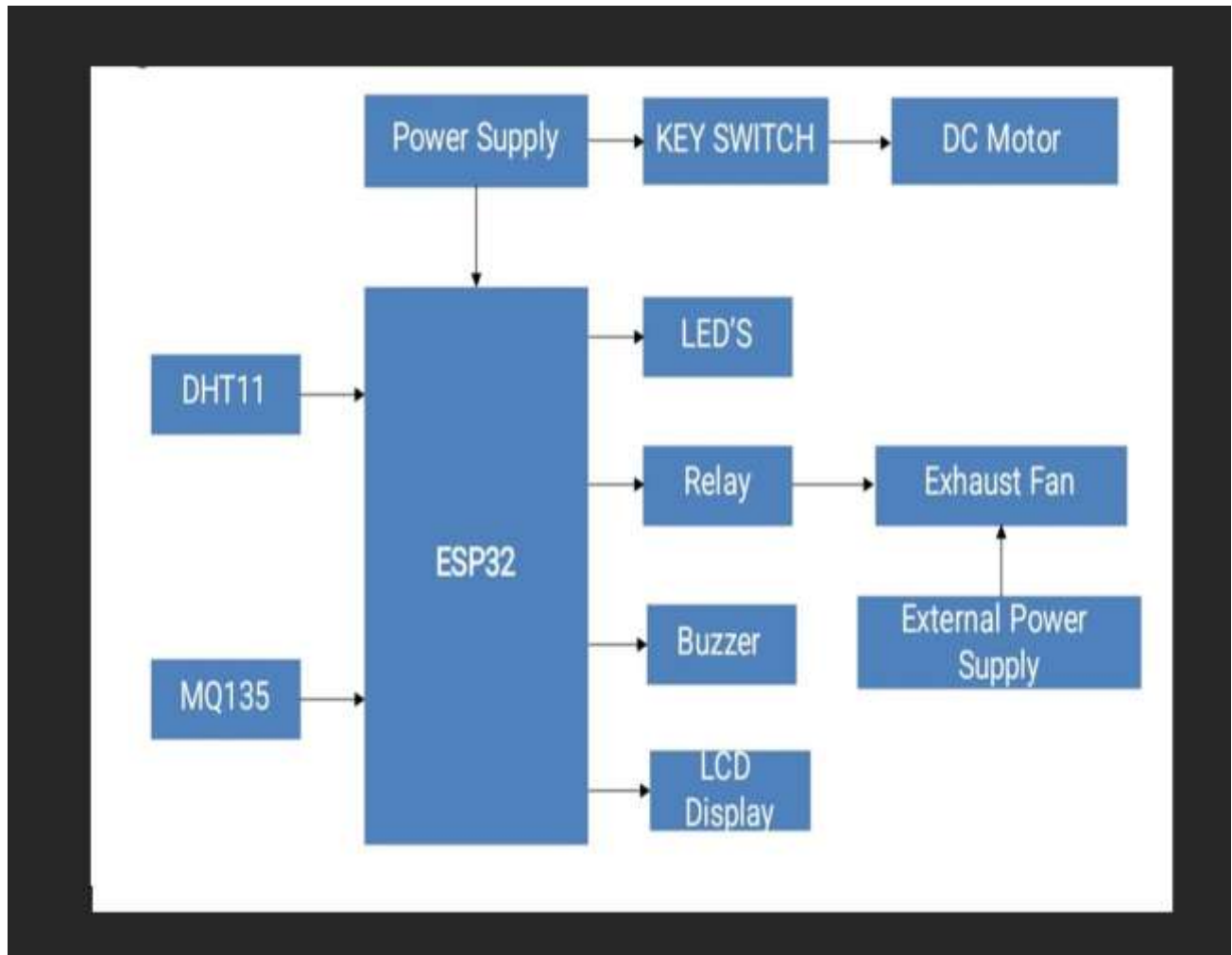


FIG 2.1 Overall block diagram of alcohol and sleep detector

Explanation:

1.Power Supply

- Provides the required voltage (usually 5V or 12V) to power the ESP32 and other components.
- Ensures stable and regulated power for proper system working.

2. DHT11 Sensor

- Measures **temperature and humidity**.
- Sends the environmental data to the ESP32 for monitoring and decision-making.

3.MQ135 Sensor

- Detects **air quality** and harmful gases like CO₂, NH₃, benzene, smoke, etc.
- Provides gas level information as an analog value to the ESP32.

4. ESP32 Microcontroller

- The **brain of the system**.
- Reads the DHT11 and MQ135 sensor data.
- Controls all output devices (LEDs, relay, buzzer, LCD).
- Makes decisions such as switching ON/OFF the fan or triggering a buzzer based on sensor values.
- Communicates with display and control modules.

5. Key Switch

- A manual ON/OFF switch used to control the **DC motor**.

6. DC Motor

- Operates based on the key switch input.
- Can be used for airflow, movement, or mechanical operation depending on system usage.

7. LEDs

- Indicate system status such as power ON, warnings, or sensor alerts.
- Controlled directly by the ESP32 digital pins.

8. Relay

- Works as an electrical switch controlled by the ESP32.
- Used to control high-power devices that cannot be driven directly by ESP32, such as exhaust fan.

9. Exhaust Fan

- Automatically turned ON/OFF based on gas level or temperature readings.
- Helps ventilate the area when harmful gases or heat are detected.

10. External Power Supply (for Exhaust Fan)

- Provides additional power to high-power devices like the exhaust fan.
- Ensures the fan does not draw heavy current from the main circuit.

11. Buzzer

- Gives an audible alarm when dangerous gas levels or high temperature/humidity are detected.
- Alerts the user about emergency conditions.

Program:

```
1  #include <DHT.h>
2  #include <LiquidCrystal_I2C.h>
3
4  #define DHTPIN 4
5  #define GAS_SENSOR_PIN 34
6  #define FAN_RELAY_PIN 16
7  #define ALARM_RELAY_PIN 17
8  #define DHTTYPE DHT11
9
10 const float TEMP_THRESHOLD_C = 31.0;
11 const int GAS_THRESHOLD = 900;
12
13 DHT dht(DHTPIN, DHTTYPE);
14 LiquidCrystal_I2C lcd(0x27, 16, 2);
15
16 void setup() {
17     Serial.begin(115200);
18     pinMode(FAN_RELAY_PIN, OUTPUT);
19     pinMode(ALARM_RELAY_PIN, OUTPUT);
20
21     digitalWrite(FAN_RELAY_PIN, HIGH);
22     digitalWrite(ALARM_RELAY_PIN, HIGH);
23 }
```

```
24     dht.begin();
25     lcd.init();
26     lcd.backlight();
27     lcd.print("Ventilation System");
28     lcd.setCursor(0, 1);
29     lcd.print("Ready!      ");
30     delay(2000);
31     lcd.clear();
32 }
33
34 void loop() {
35
36     float t = dht.readTemperature(); // Celsius
37     int gasValue = analogRead(GAS_SENSOR_PIN);
38
39     if (isnan(t)) {
40         Serial.println("Error: Failed to read from DHT sensor!");
41         lcd.clear();
42         lcd.print("T/H Read Error!");
43         delay(1000);
44         return;
45     }
46
47     bool alarmTriggered = false;
48     String triggerReason = "NONE";
```

```
50  if (t > TEMP_THRESHOLD_C) {
51      alarmTriggered = true;
52      triggerReason = "TEMP HIGH";
53  }
54
55  if (gasValue > GAS_THRESHOLD) {
56      alarmTriggered = true;
57      if (triggerReason == "NONE") {
58          triggerReason = "TOXIC GAS";
59      } else {
60          triggerReason = "BOTH";
61      }
62  }
63
64  if (alarmTriggered) {
65      digitalWrite(FAN_RELAY_PIN, LOW);
66      digitalWrite(ALARM_RELAY_PIN, LOW);
67  } else {
68      digitalWrite(FAN_RELAY_PIN, HIGH);
69      digitalWrite(ALARM_RELAY_PIN, HIGH);
70  }
71
72  Serial.print("T: "); Serial.print(t); Serial.print("°C | Gas: "); Serial.println(gasValue);
73
74
```

```
76  lcd.setCursor(0, 0);
77  lcd.print("T:");
78  lcd.print(t, 1);
79  lcd.print((char)223);
80  lcd.print("C Gas:");
81  lcd.print(gasValue);
82  lcd.print("    ");
83
84  lcd.setCursor(0, 1);
85  if (alarmTriggered) {
86      lcd.print("ALARM: ");
87      lcd.print(triggerReason);
88  } else {
89      lcd.print("Status: Normal ");
90  }
91
92  delay(2000);
93 }
```

CHAPTER 3

Chapter

3:

HARDWARE USED

3.1 Relay

- An electrically operated switch that uses an electromagnet to mechanically operate a switch.
- Allows a low-power signal from the microcontroller (e.g., 5V) to control a high-power device.



FIG 3.1 Relay

3.2 DC Motors

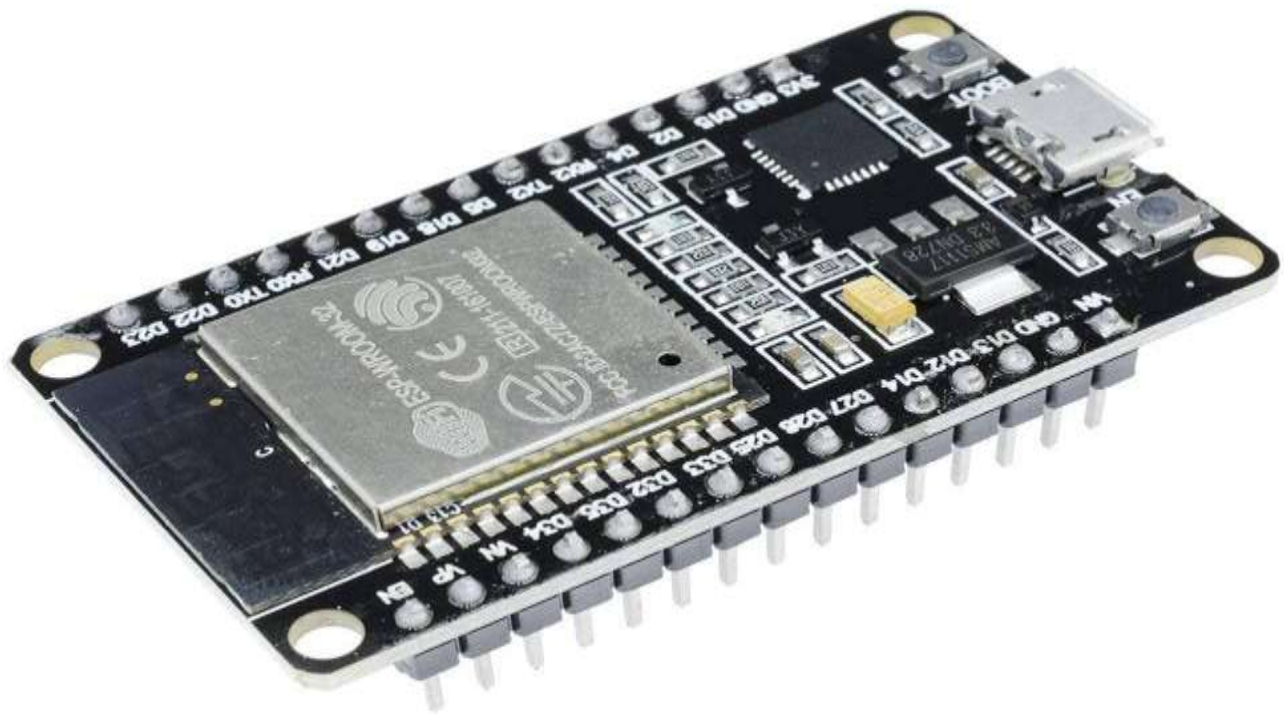
- An electric motor that converts direct current electrical energy into mechanical energy.
- Can be used to drive mechanisms like a small pump, an actuator, or another form of ventilation/movement.



FIG 3.2 DC motor

3.3ESP32

- A low-cost, low-power microcontroller with integrated Wi-Fi and Bluetooth capabilities.
- Often used as the main processing unit for IoT and embedded systems projects.



3.4. Power Supply-9V battery

- Provides the necessary electrical energy (voltage and current) to operate all the components in the system.
- Typically converts AC wall current to the required DC voltage levels (e.g., 5V or 3.3V).



FIG 3.4 (9V) Battery

3.5 MQ 135

- A gas sensor designed to detect and measure the concentration of various gases in the air.
- Commonly used for air quality monitoring as it detects gases.



FIG 3.5 MQ 135

3.6 Switch

- A simple electromechanical component used to control the flow of electricity in a circuit.
- It acts as a manual input device to turn a circuit or function ON or OFF.



FIG 3.6 Switch

3.7 DHT 11 Sensor

- A basic, low-cost digital sensor used to measure ambient environmental conditions.
- Provides readings for both temperature and humidity.



FIG 3.7 DHT11 Sensor

3.8 BUZZER

- An audio signaling device that produces a sound when current is applied to it.
- Used to provide audible alerts or notifications to the user in response to a system event.



FIG 3.8 Buzzer

3.9 LCD Display(16X2)

- A Liquid Crystal Display that presents alphanumeric characters and/or graphical information.
- Allows the system to output real-time data like sensor readings or status messages to the user.

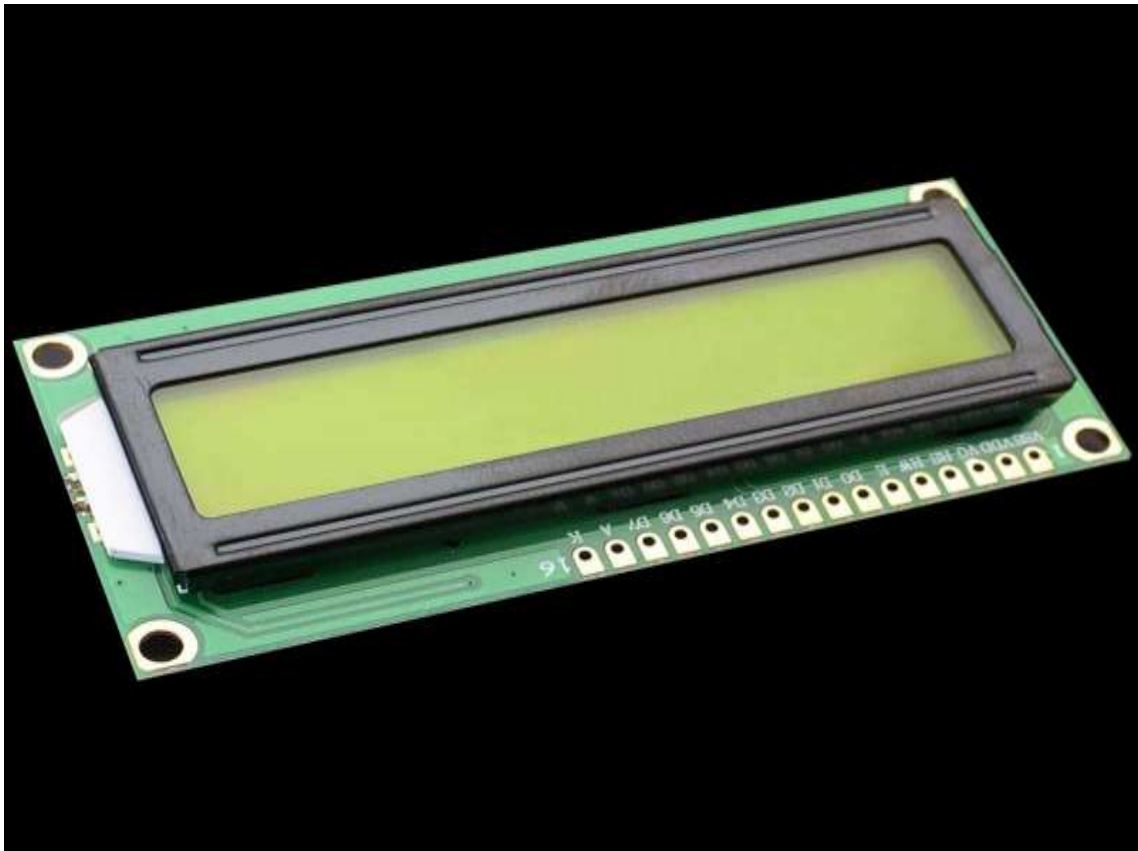


FIG 3.9 LCD Display

SOFTWARE USED

Arduino IDE:

- The Arduino Integrated Development Environment - or Arduino Software (IDE)-contains a text editor for writing code, a message area, a text console, a toolbar with buttons for common functions and a series of menus.



CHAPTER 4

4.1 APPLICATIONS

- Sericulture Farms – For large-scale silk production, ensuring healthy silkworm rearing.
- Commercial Silk Production Units – Helps small businesses and entrepreneurs improve silk yield with less manual labour.
- Climate-Controlled Storage Maintaining optimum temperature and humidity for preservation of silkworm eggs before hatching.
- Automation Integration – Monitoring systems to reduce manual intervention.

4.2 ADVANTAGES :

- Precise Environmental Control – IoT-based sensors maintain optimal temperature and humidity, ensuring healthy silkworm growth.
- Reduced Human Error – Automation minimizes mistakes in monitoring and controlling the incubator compared to manual methods.
- Data Recording & Analysis – IoT systems store temperature, humidity, and growth data, helping improve future batches.
- Improved Productivity & Quality – By maintaining ideal conditions, silkworms produce better quality silk and in higher quantity.

4.3 DISADVANTAGES:

1. High Initial Cost

- Purchasing incubators, heaters, sensors, and controllers can be expensive.
- Not affordable for small-scale farmers or rural areas.

2. Requires Continuous Power Supply

- Incubators need stable electricity for temperature and humidity control.
- Power cuts can disturb larval growth and reduce cocoon quality.

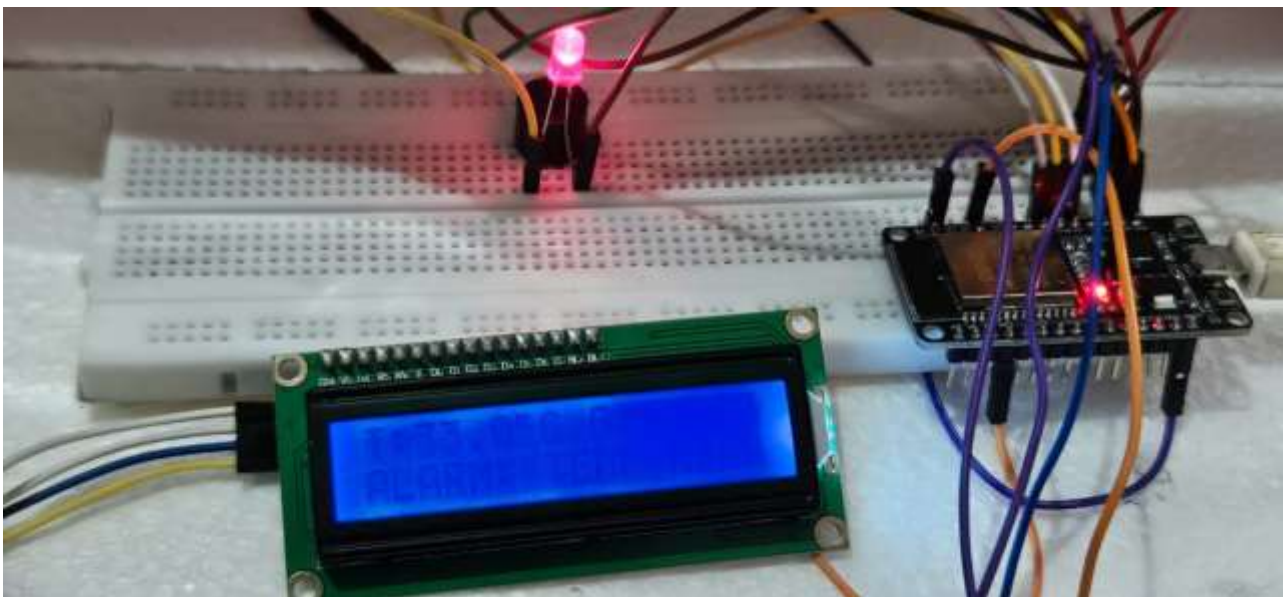
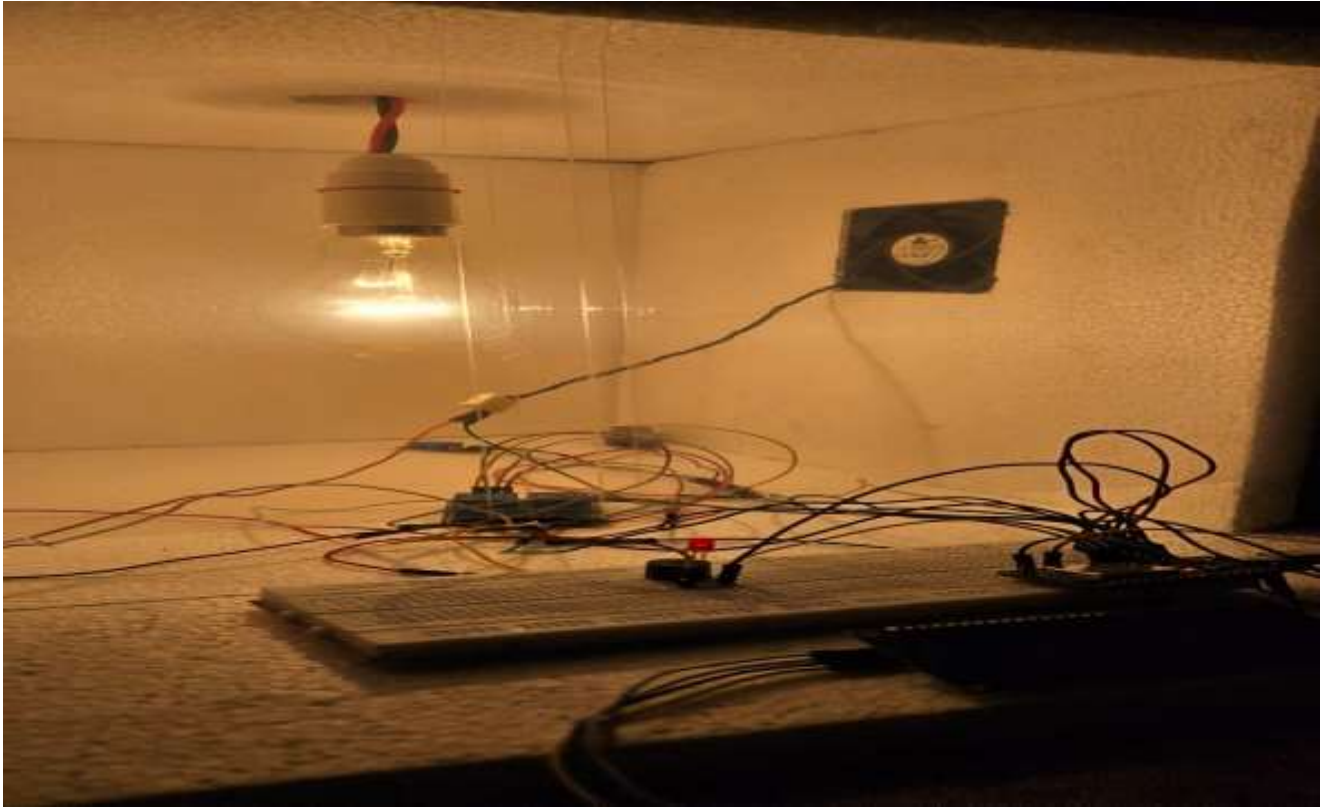
3. Maintenance & Calibration Needed

- Sensors (temperature/humidity) must be regularly checked and calibrated.
- Fans, heaters, and controllers require periodic maintenance.

CHAPTER 5

RESULTS AND CONCLUSION :

RESULT:



Future Scope:



1. Connectivity & IoT

Cloud Integration: Utilize the ESP32's Wi-Fi to log data to a cloud platform (e.g., ThingSpeak) for remote monitoring, data analytics, and historical logging.

Remote Alerts: Develop a mobile application for users to receive push notifications and remotely control the Exhaust Fan.

2. Performance & Control

Sensor Upgrade: Replace the DHT11 with the BME280 for higher accuracy (adding pressure data).

Proportional Control: Implement Variable Speed Control (PWM) for the Exhaust Fan to adjust its speed proportionally to the severity of the air quality or temperature level, rather than simple ON/OFF control.

3. Smart Automation

Predictive Management: Incorporate historical data analysis to predict environmental changes and pre-emptively activate ventilation systems.

Conclusion:

This project successfully developed a Smart Environmental Monitoring and Control System using the ESP32 microcontroller, integrating DHT11 (temperature/humidity) and MQ135 (air quality) sensors.

The system achieves real-time environmental control by using the sensor data to automatically manage the Exhaust Fan (via Relay) for ventilation and activating the Buzzer and LEDs for alerts. The LCD Display provides immediate local feedback. The design demonstrates a robust, autonomous, and scalable solution for environmental management in enclosed spaces.

CHAPTER 6

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